Case 4: Historical Rates of Appreciation of Residential Real Estate Assets in Dade County, Florida

February 27, 2026

1 Background

In the US, residential real estate has historically been the dominant asset held by the individual investor: as a fraction of total household wealth, home equity has dominated savings, stocks and bonds and other assets. In gross, real estate plays a significant role in the economy. For this reason, there is intense interest in monitoring the behavior of this asset class across scales, from the macroeconomy to local markets to individual properties. Real estate indices provide one tool for this purpose. Broadly defined, a real estate index is an estimate of the rate of appreciation – the change in market value per unit time – of a typical home within a geographically defined market. They are used by institutional investors to judge risk in a market and to measure market performance. They are also used as an aid to property appraisal and valuation.

Real estate valuations vary significantly across time and space: 'booms' and 'busts' do not occur simultaneously across the country. Nor do external factors affect all market segments equally. The 'top' and 'bottom' of a market may respond to prevailing economic conditions in very different ways.

In addition, sale prices of homes sold at any given time in most geographical regions are extremely variable as a result of differences in quality (upkeep, lot size, age, number of bedrooms and baths, and other tangible and intangible features). Yet it is necessary to draw accurate estimates of market appreciation for such regions from available sales data.

One approach to this source of variation has been to account for the quality of each home observed sold. 'Hedonic price indices' attempt to do this. Hedonic variables measure the quality of homes sold – their size, age, and neighborhood characteristics – but these are often not observable or obtainable. In situations in which they are available, there may be disagreement over which are appropriate and how to use them to modify the index. Another shortcoming of hedonic models is that they are not broadly applicable: Variables appropriate or available for one market may not be so in another.

One type of index that avoids the problems inherent to hedonic models utilizes repeat–sales data, data consisting of consecutive sales of homes. The advantage to modeling repeat sales on housing units is that each unit acts, in a way, as its own covariate. Hence, as long as home quality is constant between sales, it is not necessary to explicitly account for it. However violations of this assumption may lead to bias, increased variance, or both in the estimated index.

2 Data

The current data set was obtained from the Dade County, Florida Department of Property Appraisal late 1995. It comprises selected fields drawn from 268,879 residential single family folios (records corresponding to a property). Sales were first electronically recorded in 1970, and the data are current through mid–1995.

Only owner–occupied, single–family dwellings are included here, while homes that were known to have been modified and new homes (for which resales are clearly not available) were excluded from the analysis. Only the three most recent sales for each home are recorded by the appraiser's office (regions where there has been high ownership turnover may, as a result, be underrepresented in early time periods, but this

aspect of the data will be ignored). Only records of sales known to have been conducted at arm's length were used. Examples of sales not considered to be at arm's length are those conducted between related parties, foreclosures, multifolio sales, those that represent unconventional financial terms, sales based on preconstruction prices, and sales that otherwise do not fit the market (as judged by the property appraiser's office).

Repeat sales are defined as any pair of arm's length sales for a given home. A home may have zero, one, or two repeat sales according to the number of arm's length sales recorded for the property.

The data file `dadeRE.RData` contains 77,493 records with dates of first and second sale between January 1, 1971, and December 31, 1995 and the following variables:

outputID Folio or property ID (characters 1:13), repeat sale ID valued 'a' or 'b' (character 14).

zipcode Postal code in which the property is situated.

amount1, amount2 First and second sale prices received for the property.

month1, day1, year1 Month, day and year of the first sale. Note that day is meaningless (it is always =15).

month2, day2, year2 Month, day and year of the second sale. Note that day is meaningless (it is always =15).

xfeat Bonus mystery variable!

beds Number of bedrooms in the dwelling.

baths Number of full or 3/4 baths in the dwelling.

hbath Number of half–baths in the dwelling.

floors Number of floors in the dwelling

lotsqft Square footage of the lot on which the dwelling is situated.

muni Municipality ID. Coding currently unknown.

bldgsqft Square footage of the dwelling.

propertyiD, pairID see description of `outputID` above.

strata Grouping of homes into size–based strata. See Below.

- **S1** Small homes. Those with only one bedroom or two bedrooms with one full bath.
- **S2** Modest homes. Two bedroom homes with two or more full bathrooms or three bedrooms with only one full bath.
- **S3** The mode. Three bedrooms, two full baths.
- **S4** Large homes. Three beds, three or more full baths; four beds, one or two baths.
- **S5** Very large homes. Four beds, three or more baths; five or more beds.

The data file `DadeSegs` contains line segments that can be used to construct a contemporaneous zipcode map of the county (see Figure 2; code in `PlotMap.R`). Note that zipcodes and their boundaries change over time; this region has changed considerably over the past 30 years.

3 Objectives

Your group will be assigned a stratum to focus on.

1. Describe zipcode level geographic variation in relative abundance of homes in your stratum.
2. Describe sales activity by quarter.
3. Estimate the quarterly rates of appreciation experienced by a typical single family home in that stratum for the period begining January 1, 1971 and ending December 31, 1995.
4. Does temporal variation in sales activity and/or rates of appreciation correlate with interest rates on 30 year fixed rate mortgages? If so, how? See Figure 1.

4 Complicating Factors

Hurricane Andrew (`https://en.wikipedia.org/wiki/Hurricane_Andrew`), a very strong category 5 storm, passed over the southern portion of the county in August 1992. Property damage was extreme. Quoting the Wikipedia entry: "the hurricane stripped many homes of all but their concrete foundations and caused catastrophic damage. In total, Andrew destroyed more than 63,500 houses, damaged more than 124,000 others, caused $27.3 billion in damage (equivalent to $65 billion in 2023) and left 65 people dead." Needless to say, many repeat sales differences spanning this event reflect this damage.

5 Assignment

- Conduct an exploratory data analysis for class on **Monday March 16th**. Begin preparing your project slide deck by describing this analysis, highlighting any issues you encountered and unresolved questions you may have, and summarizing your preliminary inferences and discoveries. Be prepared to share these in class.
- Add to your project slides *about* five slides describing your final Bayesian model(s) and inferences, highlighting assumptions and shortcomings. Turn in your full set of slides (EDA + model–based analysis) and be prepared to present these in class **Monday March 23rd**.
- Describe your final results in a formal report of up to four pages (writing and display equations); all tables and figures should appear in an appendix and be referenced by name in the main text. Your report should highlight all relevant aspects of your analysis (exploratory and modeling) and include graphical and numerical summaries that aid in communicating your results. You may also include code and other supporting material in the supplemental appendix. Due in class **Wednesday March 25th**.

Figure 1: The Federal Reserve Bank of St. Louis' Economic Data dashboard displaying a time series of the interest rate charged for 30 year fixed rate mortgages. Source: `https://fred.stlouisfed.org/series/MORTGAGE30US`

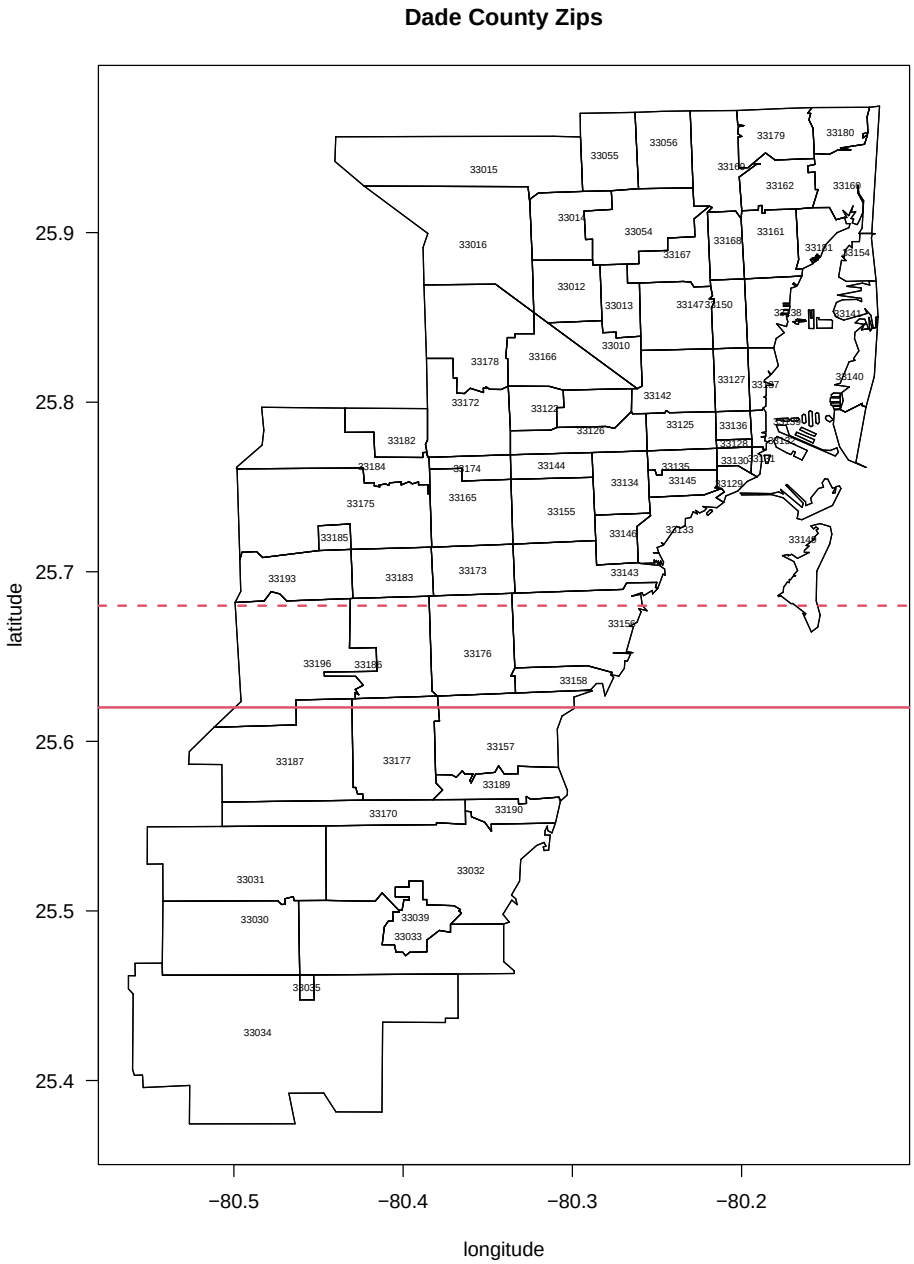

Figure 2: Map of Dade County, FL zipcode boundaries circa 1995. Solid horozontal red line roughly indicates path of Hurricane Andrew's center of circulation; dashed red line indicates the northern extent of the storm's eye wall (see Figure 3).

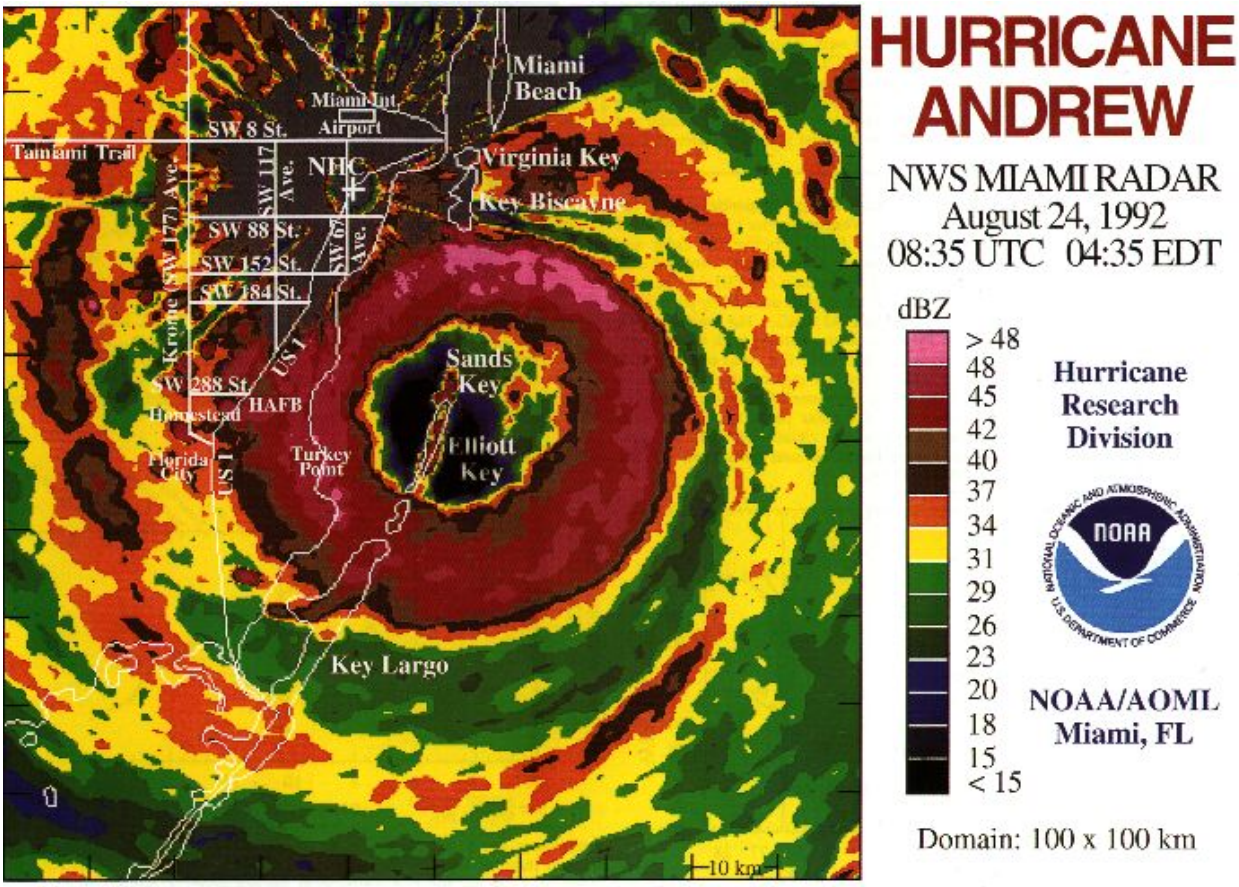

Figure 3: Last available radar image of Hurricane Andrew prior to landfall, providing geographic reference points. Source: Wikipedia/NWS.